Coldspots Analysis Report

Issue #1: Visualizing coldspots in genes using ChimeraX | Gadis | October 14, 2026

Question

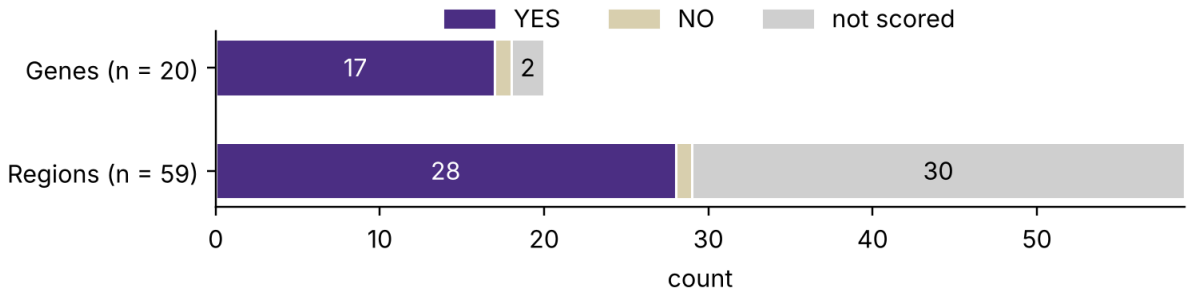
Do regional coldspots fall on intrinsically disordered regions (IDRs)? These are parts of a protein that don't fold into a fixed shape. I used low AlphaFold confidence (pLDDT) outside a folded domain as the sign of disorder, following Fawzy & Marsh 2025.

What I did

- Randomly picked 20 genes from the regional coldspots file.
- Looked up each protein's UniProt ID and loaded its AlphaFold model in ChimeraX, colored by pLDDT.
- Highlighted each coldspot (amino acid start to end) and checked where it landed.
- Scored each coldspot YES if it sat on a low confidence, flexible region, NO if it sat on a folded region, and left it unscored if there was no usable model.

Results

Level	YES	NO	Not scored	YES out of scored
Genes	17	1 (TP53)	2 (KMT2B, ALMS1)	17 of 18
Coldspot regions	28	1	30	28 of 29



Every gene I could score had its coldspots sitting on disordered regions. The only NO was TP53, but the structure I used had the wrong UniProt ID (Q9Y2B4 instead of P04637), so I was probably looking at a different protein. TP53's coldspots are near the ends of the protein, which are the parts of p53 known to be floppy, so I expect it to become a YES once I reload it. KMT2B and ALMS1 couldn't be scored because they're too long for one AlphaFold model, and together they hold about half of the coldspot regions in my sample.

What this means

The result supports the hypothesis: coldspots seem to land on flexible, disordered parts of proteins. This makes sense biologically, since a single amino acid change is less likely to break a region that has no fixed shape than a folded domain.

Limitations

- **No chance comparison yet.** Some of these proteins are mostly disordered, so a coldspot could land on a disordered part just by luck. Comparing each coldspot to random spots in the same protein would show if the overlap is more than expected.
- **Small sample.** 20 genes, and only 18 could be scored.
- **Scoring by eye.** Calls were made by looking at colors in ChimeraX, not by measuring the exact share of residues below the pLDDT cutoff.
- **Missing long proteins.** Very long proteins have no single AlphaFold model, so they are left out by design, which could bias the sample.
- **Isoforms.** Coldspot positions come from one transcript, and AlphaFold uses the canonical UniProt sequence. If they differ, residue numbers may not line up.

Next steps

- Reload TP53 with the correct UniProt ID (P04637) and rescore.
- Score KMT2B and ALMS1 using AlphaFold fragment models or a disorder database like MobiDB.
- Run the random placement test and expand to more genes.